Integration: Reversing the Chain Rule Part 2

Questions 1–37

Question 1. $f'(x)e^{f(x)}$

Find

a) $\int e^{2x+1}\,dx$

b) $\int e^{5-3x}\,dx$

c) $\int xe^{x^2}\,dx$

d) $\int 4x^3e^{6x^4}\,dx$

Question 2. Find

a) $\int \cos x\, e^{\sin x}\,dx$

b) $\int \sin 3x\, e^{6\cos 3x}\,dx$

c) $\int 4\cos\frac{1}{2}x\, e^{\sin\frac{1}{2}x}\,dx$

d) $\int 9\sin\frac{1}{3}x\, e^{3\cos\frac{1}{3}x}\,dx$

Question 3. Find

a) $\displaystyle\int (\ln 2) 2^x e^{2^x}\, dx$

b) $\displaystyle\int 3^x e^{3^x}\, dx$

c) $\displaystyle\int \frac{1}{\ln 5} 5^x e^{5^x+1}\, dx$

d) $\displaystyle\int 4^x e^{4^{x+1}}\, dx$

Question 4. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \sec^2 x\, e^{\tan x}$

b) $\frac{dy}{dx} = 2 \operatorname{cosec} x \cot x\, e^{4 \operatorname{cosec} x}$

c) $\frac{dy}{dx} = 3 \sec 6x \tan 6x\, e^{\sec 6x}$

d) $\frac{dy}{dx} = -8 \operatorname{cosec}^2 \frac{1}{2} x\, e^{16 \cot \frac{1}{2} x}$

Question 5. The curve y has gradient function

$$\frac{dy}{dx} = -\frac{e^{1/x}}{x^2}$$

and passes through the point $(1, -e)$. Write y in terms of x.

Question 6. Evaluate the following definite integrals:

a) $\displaystyle\int_0^1 x^2 e^{x^3}\, dx$

b) $\displaystyle\int_0^{\pi/4} \cos 2x\, e^{\sin 2x}\, dx$

c) $\displaystyle\int_1^2 (\ln 8) 2^x e^{2^x}\, dx$

d) $\displaystyle\int_0^{\pi/16} 60 \sec^2 4x\, e^{\tan 4x}\, dx$

Question 7. Find the exact value of

$$\lim_{\delta x \to 0} \sum_{x=1}^{4} \frac{e^{2\sqrt{x}}}{\sqrt{x}} \delta x$$

Question 8. $f'(x)\sin(f(x))$ **or** $f'(x)\cos(f(x))$

Find

a) $\displaystyle\int \sin\left(5x + \frac{\pi}{7}\right) dx$

b) $\displaystyle\int 8x \cos 2x^2 \, dx$

c) $\displaystyle\int (2x+1)\sin(x^2 + x - 1)\, dx$

d) $\displaystyle\int (x^2 - 3)\cos\left(6x - \frac{2}{3}x^3\right) dx$

Question 9. Find

a) $\int \cos x \sin(\sin x)\, dx$

b) $\int \sin x \sin(\cos x)\, dx$

c) $\int \cos x \cos(\sin x)\, dx$

d) $\int \sin x \cos(\cos x)\, dx$

Question 10. Find

a) $\int (\ln 5) 5^x \sin(5^x)\, dx$

b) $\int 9^x \cos(9^x)\, dx$

c) $\int \dfrac{\sin(\ln x)}{x}\, dx$

d) $\int \dfrac{\cos(2 \ln x)}{x}\, dx$

Question 11. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \operatorname{cosec}^2 2x \sin(\cot 2x)$

b) $\frac{dy}{dx} = 4 \operatorname{cosec} 3x \cot 3x \cos(8 \operatorname{cosec} 3x)$

c) $\frac{dy}{dx} = 9 \sec^2 \frac{1}{3}x \sin(90 \tan \frac{1}{3}x)$

d) $\frac{dy}{dx} = \sec \frac{2}{3}x \tan \frac{2}{3}x \cos(18 \sec \frac{2}{3}x)$

Question 12. The curve y has gradient function

$$\frac{dy}{dx} = 12x \sin x^2$$

and passes through the point $\left(\frac{\sqrt{\pi}}{2}, 24\right)$. Write y in terms of x.

Question 13. Evaluate the following definite integrals:

a) $\displaystyle\int_{\ln(\pi/2)}^{\ln \pi} e^x \sin(e^x)\, dx$

b) $\displaystyle\int_{0}^{\sqrt{\pi}/4} 40x \cos 8x^2\, dx$

c) $\displaystyle\int_{\sqrt{2\pi}}^{\sqrt{3\pi}} 18x \sin x^2\, dx$

d) $\displaystyle\int_{\pi/4}^{\pi/3} (24x - 7\pi) \cos(12x^2 - 7\pi x + \pi^2)\, dx$

Question 14. Find the exact value of

$$\lim_{\delta x \to 0} \sum_{x=-16/\pi}^{8/\pi} \frac{\sin(4/x)}{8x^2} \delta x$$

Question 15. $\dfrac{f'(x)}{f(x)}$

Find

a) $\displaystyle\int \frac{1}{x+1}\,dx$

b) $\displaystyle\int \frac{2}{2x-5}\,dx$

c) $\displaystyle\int \frac{8}{4x+3}\,dx$

d) $\displaystyle\int \frac{9}{5x-7}\,dx$

e) $\displaystyle\int \frac{3}{1-4x}\,dx$

f) $\displaystyle\int \frac{15}{5-3x}\,dx$

Question 16. Find

a) $\displaystyle\int \frac{2x}{x^2+3}\,dx$

b) $\displaystyle\int \frac{3x^2}{5+x^3}\,dx$

c) $\displaystyle\int \frac{15x}{5x^2-4}\,dx$

d) $\displaystyle\int \frac{10x^3}{6-x^4}\,dx$

Question 17. Find

a) $\displaystyle\int \frac{2x+4}{x^2+4x-3}\,dx$

b) $\displaystyle\int \frac{3x^2-3}{x^3-3x+9}\,dx$

c) $\displaystyle\int \frac{x+1}{4x^2+8x+5}\,dx$

d) $\displaystyle\int \frac{6x^2-24x}{x^3-6x^2-1}\,dx$

Question 18. Find

a) $\displaystyle\int \frac{e^x}{e^x+1}\,dx$

b) $\displaystyle\int \frac{(\ln 2)2^x-1}{2^x-x}\,dx$

c) $\displaystyle\int \tan x\,dx$

d) $\displaystyle\int \cot x\,dx$

Question 19. Find

a) $\displaystyle\int \frac{\cos 2x}{\sin 2x + 1}\, dx$

b) $\displaystyle\int \frac{6\sin 3x}{4 - \cos 3x}\, dx$

c) $\displaystyle\int \frac{24\cos 6x}{3\sin 6x - 5}\, dx$

d) $\displaystyle\int \frac{18\sin 9x}{15\cos 9x + 4}\, dx$

Question 20. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \frac{(\ln 3)3^x + e^x}{3^x + e^x}$

b) $\frac{dy}{dx} = \frac{8 + 4\sec^2 2x}{4x + \tan 2x}$

c) $\frac{dy}{dx} = \frac{\cos x + \frac{1}{2\sqrt{x}}}{\sin x + \sqrt{x}}$

d) $\frac{dy}{dx} = \frac{30e^{3x} - 20e^{2x}}{e^{2x} - e^{3x}}$

Question 21. The curve y has gradient function

$$\frac{dy}{dx} = \frac{2x+1}{(x-1)(x+2)}$$

and passes through the point $(2, 2\ln 2)$. Write y in terms of x.

Question 22. Evaluate the following definite integrals:

a) $\displaystyle\int_0^1 \frac{12}{3x+1}\,dx$

b) $\displaystyle\int_{\sqrt{2}}^{\sqrt{3}} \frac{3x}{x^2-1}\,dx$

c) $\displaystyle\int_{\pi}^{2\pi} \frac{\sin x}{\cos x + 2}\,dx$

d) $\displaystyle\int_{\ln 2}^{\ln 4} \frac{4e^x}{4+e^x}\,dx$

Question 23. Find the exact value of

$$\lim_{\delta x \to 0} \sum_{x=1}^{3} \frac{6x+1}{6x^2+2x+1} \delta x$$

Question 24. $f'(x)(f(x))^n$

Find

a) $\int e^x (e^x + 5)^3 \, dx$

b) $\int e^{3x} (e^{3x} - 3)^4 \, dx$

c) $\int 2e^{4x} (4e^{4x} - 1)^5 \, dx$

d) $\int 10e^{x/2} \sqrt{e^{x/2} + 2} \, dx$

Question 25. Find

a) $\int \cos x \sin^2 x \, dx$

b) $\int \cos x \sin^3 x \, dx$

c) $\int \sin x \cos^4 x \, dx$

d) $\int \sin 2x \cos^5 2x \, dx$

Question 26. Find

a) $\int \sec^2 x(\tan x - 4)^5 \, dx$

b) $\int \sec^2 3x \tan^3 3x \, dx$

c) $\int 2^x(2^x + 4)^4 \, dx$

d) $\int 3^{x+1}(3^x - 2)^5 \, dx$

Question 27. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \frac{(\ln x+3)^3}{x}$

b) $\frac{dy}{dx} = \frac{(2\ln x+5)^5}{12x}$

c) $\frac{dy}{dx} = \sec x \tan x \,(\sec x + 2)^3$

d) $\frac{dy}{dx} = \operatorname{cosec}^8 x \cot x$

Question 28. Evaluate the following definite integrals:

a) $\displaystyle\int_{\ln 2}^{\ln 4} e^x(4e^x - 5)^3 \, dx$

b) $\displaystyle\int_0^{\pi/2} \cos x \sin^4 x \, dx$

c) $\displaystyle\int_{2\pi}^{3\pi} \sin \frac{1}{2}x \cos^5 \frac{1}{2}x \, dx$

d) $\displaystyle\int_e^{e^2} \frac{(\ln x + 3)^5}{2x} \, dx$

Question 29. Mix

Given that k is a positive constant, find

$$\int_{k}^{2k} \frac{8}{4x+k}\,dx$$

Question 30. Given that k is a positive constant, find

$$\int_{1/k}^{2/k} \frac{k}{kx+5}\,dx$$

Question 31. Given that k is a positive constant, find

$$\int_{2/k}^{3/k} \frac{k^2 x}{k^2 x^2 - 2}\, dx$$

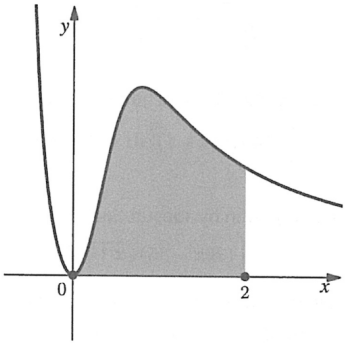
Question 32. Given that k is a positive constant, show that

$$\int_{\sqrt{k}}^{\sqrt{4k}} 8xe^{x^2}\, dx = 4e^k(e^{3k} - 1)$$

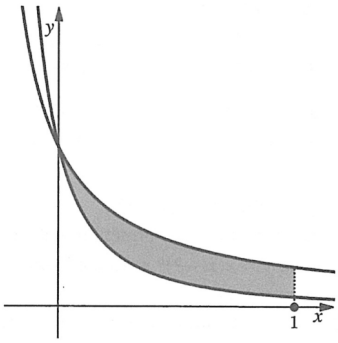
Question 33. Find the exact value of

$$\int_{\ln(\sqrt{\pi}/4)}^{\ln(\sqrt{\pi}/2)} 8e^{2x}\sin(4e^{2x})\,dx$$

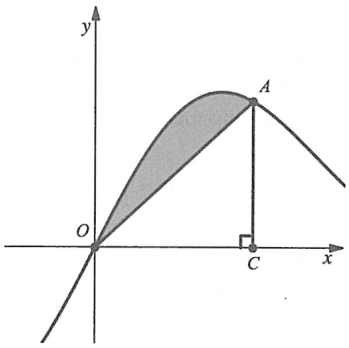
Question 34. The diagram below shows the curve with equation $y = \dfrac{2x^2}{4x^3+1}$. Find the exact value of the shaded area.



Question 35. The diagram below shows the curves $y = \dfrac{3}{3x+1}$ and $y = \dfrac{3}{(3x+1)^2}$. Find the exact shaded area between the two curves.



Question 36. The diagram below shows the curve with equation $y = \sin 2x \cos^5 2x$ where C has x-coordinate $\dfrac{\pi}{12}$. Find the exact shaded area.



Question 37. Prerequisite knowledge for Q37: Quotient Rule

The diagram below shows $y = \dfrac{\sin 3x}{\cos 3x + 2}$. A and C are turning points on the curve, and the angle ABC is 90°. Find the exact shaded area.

